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Autoregressive Sequential Pretraining for Visual Tracking

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https://ARPTrack.github.io

Abstract

*Recent advancements in visual object tracking have shifted towards a sequential generation paradigm, where object deformation and motion exhibit strong temporal dependencies. Despite the importance of these dependencies, widely adopted image-level pretrained backbones barely capture the dynamics in the consecutive video, which is the essence of tracking. Thus, we propose **A**uto**R**egressive Sequential **P**retraining (ARP), an unsupervised spatio-temporal learner, via generating the evolution of object appearance and motion in video sequences. Our method leverages a diffusion model to autoregressively generate the future frame appearance, conditioned on historical embeddings extracted by a general encoder. Furthermore, to ensure trajectory coherence, the same encoder is employed to learn trajectory consistency by generating coordinate sequences in a reverse autoregressive fashion, a process we term backtracking. Further, we integrate the pretrained ARP into ARTrackV2, creating ARPTrack, which is further fine-tuned for tracking tasks. ARPTrack achieves state-of-the-art performance across multiple benchmarks, becoming the first tracker to surpass 80% AO on GOT-10k, while maintaining high efficiency. These results demonstrate the effectiveness of our approach in capturing temporal dependencies for continuous video tracking. The code will be released soon.*

1. Introduction

Recent advances in Visual Object Tracking [3, 36, 38, 42, 48], have witnessed a paradigm shift from traditional per-frame template matching [2, 23, 29, 41, 44, 56, 66] to sequential generation [1, 14, 59], which has set new performance benchmarks, pioneered by ARTrack [59]. This transformation aligns with the inherently temporal nature of tracking [6, 39, 62, 70], where object deformation and motion exhibit strong sequential dependencies. This advancement has far-reaching implications for time-critical applications, spanning from embodied intelligence, autonomous driving, human-computer interaction, and augmented reality systems.

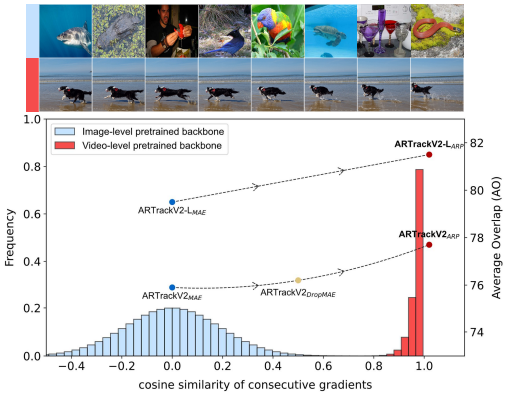


Figure 1. **Comparison between image-level and video-level pretrained backbones**. **Top**: Representative training samples from image-based datasets (blue) and video-based datasets (red). **Bottom**: (1) The gradient cosine similarity between consecutive iterations follows a normal distribution during image-level pretraining, whereas it demonstrates significant temporal correlation in the video-level pretrained backbone, and (2) when fine-tuned with ARTrackV2, the video-level pretrained approach (red) demonstrates superior tracking performance on GOT-10k then image-level (blue) and pairs-level (yellow).

Moreover, the paradigm shift offers a novel perspective on conventional approaches. As demonstrated in Figure 1, the gradient patterns between image-level and video-level pretrained models exhibit stark disparities, with video models demonstrating superior perception to temporal dynamics. This reveals a fundamental constraint in current tracking frameworks: despite their widespread adoption, image-

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level pretrained backbones [15, 44, 66] inherently lack the capacity for temporal modeling, failing to capture continuous appearance variations and trajectory evolution. Although recent attempts like DropMAE [60] introduce temporal modeling through spatial-attention dropout between frame pairs, such discrete two-frame interactions are insufficient for capturing long-range successive temporal dependencies in real-world tracking scenarios, which often span thousands of frames with continuous object deformation and motion.

Inspired by the success of Generative Pretraining [5, 22, 50, 51, 55] in language modeling, which has demonstrated remarkable effectiveness in capturing contextual dependencies within large language models, we propose **A**uto**R**egressive **P**retraining (ARP). This unsupervised approach is specifically designed to learn to capture both appearance variations and trajectory evolution in continuous video sequences, as an avenue to understanding sequential dependencies [11] which tracking needs.

Our method begins by sampling continuous video clips to maintain temporal consistency and randomly initializing the target state (appearance and coordinates) from the first frame. At the core of our approach is training a Vision Transformer (ViT) backbone, which participates in two essential components: appearance prediction and trajectory verification. First, the ViT acts as an appearance encoder that leverages a conditional diffusion model for autoregressive appearance generation. For each video clip, it sequentially processes frames and incorporates historical appearance embeddings to guide the diffusion model in predicting the target's appearance in subsequent frames. This mechanism allows the model to capture continuous appearance variations and learn temporal evolution patterns. However, modeling appearance evolution alone is insufficient, as it neglects essential trajectory physics crucial for robust tracking, especially in challenging scenarios with occlusions and distractors. To address this limitation, we use the same ViT architecture to ensure trajectory consistency by performing back-tracking verification through autoregressive trajectory sequence generation. Our approach treats the initialized coordinates of the target as labels and employs predicted appearances as templates, while using historical trajectory sequences as prompts, the same as ARTrackV2, to generatively back-tracking.

To validate our pretraining approach, we integrated our pretrained model into the ARTrackV2 architecture, creating ARPTrack, which emphasizes temporal dependencies. Our evaluations show that ARPTrack achieves state-of-the-art performance, reaching an AO score of 81.5% on the GOT-10k benchmark, the first tracker surpassing 80% AO. ARPTrack also maintains competitive performance on LaSOT and TrackingNet datasets while preserving the computational efficiency of ARTrackV2. Furthermore, as demonstrated in Figure 1, our continuous video-pretrained backbone shows significant improvements over both image-level [32] and pairs-level [30, 60] pretrained backbones. These results confirm the effectiveness of our approach in capturing temporal features, opening new possibilities for advanced video understanding tasks.

2. Related Work

Tracking Paradigm. Traditional visual object tracking [18, 20, 71] has predominantly relied on a frame-by-frame template matching paradigm [2, 23, 27, 29, 33, 41, 46, 56, 58], where a reference template is used to locate the target within each frame's search region. These approaches typically employ a backbone network for feature extraction [9, 15, 44, 66], followed by multiple specialized heads [4, 13, 16, 26, 40, 52, 53, 63, 69] for subtasks such as scale estimation and center point localization. This design, while functional, involves complex post-processing [19, 54] and neglects crucial temporal dependencies [4, 20, 43, 45, 57, 64, 65, 67] inherent in continuous tracking.

Recent advances have witnessed a paradigm shift towards sequential generation [1, 14, 59], establishing new performance benchmarks in the field. This transformation aligns with the temporal nature of tracking, where object deformation and motion exhibit strong sequential patterns. SeqTrack [14] introduces a streamlined encoder-decoder architecture that directly generates bounding box coordinate sequences, eliminating the need for complex post-processing and manually designed task-specific heads. Beyond that, ARTrack [59] reconceptualized tracking as coordinate sequence interpretation, enabling direct trajectory estimation through an autoregressive model guided by historical motion patterns. Building upon ARTrack, ARTrackV2 [1] advanced this approach by proposing a unified framework that seamlessly integrates both trajectory and appearance evolution in an autoregressive manner.

Despite the promising perspective offered by the sequential generation paradigm, current tracking systems still heavily rely on image-level pre-trained backbones [15, 17, 66], which inherently lack temporal modeling capabilities. In this paper, we propose a video-level pretraining approach designed to capture both deformation patterns and trajectory evolution of targets in consecutive video sequences.

Unsupervised Pretraining. Recent advances in unsupervised pretraining have greatly improved performance across various tasks. In computer vision, MoCo [31] reframed contrastive learning as a dictionary look-up task, using a dynamic memory bank and a momentum-updated encoder. MAE [32] introduced masked image modeling, reconstructing masked patches from visible ones, showing strong transfer capabilities. Building on this, DropMAE [60] added temporal modeling through spatial-attention dropout be-

tween frames, specifically for object tracking. However, these methods still lack explicit long-term temporal modeling, which is crucial for tracking. In contrast, generative pretraining in NLP [5, 22, 50, 51, 55], with its strong contextual learning and reasoning abilities, offers valuable insights on how to model temporal reasoning [10]. Inspired by this, we propose Autoregressive Sequential Pretraining for tracking. Our approach uses a diffusion model [34, 68] to generate the next-frame appearance of targets and autoregressively predicts the next coordinates in the target's trajectory, effectively capturing temporal dependencies.

3. Autoregressive Sequential Pretraining

We propose **A**utore**R**ressive **P**retraining (ARP), an unsupervised learning framework that models both object deformation and motion dynamics in continuous videos. As shown in Figure 2, ARP builds upon Vision Transformer (ViT) [21] encoding and Conditional Diffusion modeling, operating through two key mechanisms: appearance prediction and trajectory verification.

3.1. Appearance Prediction

Given continuous video clip sampling from successive videos, we randomly initialize the target state including the bounding box and appearance in the first frame. Then we progressively model appearance variations through time-continuous, formulated as:

$$P\left(\hat{\boldsymbol{X}}^{t+1} \middle| \underbrace{\overrightarrow{\boldsymbol{Z}^{t-N:t-1}}}_{\text{historical appearance}}, \boldsymbol{X}^{t}\right) \tag{1}$$

where $\hat{\boldsymbol{X}}^{t+1}$ indicates the predicted next-frame appearance, $\boldsymbol{Z}^{t-N:t-1}$ represent the historical appearance context spanning the previous N frames, $\rightarrow$ represent the sequence order (from $t-N$ to $t-1$), while $\boldsymbol{X}^{t}$ denotes the search image at time step t.

The autoregressive appearance prediction operates in two stages: temporal feature extraction and conditional diffusion generation. In the feature extraction stage, the historical appearance and the current search image are divided into patches, flattened, and projected. These patches, added with positional, identity and temporal embeddings, form sequences of appearance and search embeddings. The ViT encoder then processes all embeddings parallelly into temporal feature representations. In the generation stage, we employ a conditional diffusion model. Different from traditional conditional diffusion [34, 68] which injects Gaussian noise as input, we leverage search embeddings as the noise source. During the iterative denoising process, historical appearance embeddings as conditions guide the generation of the target's next-frame appearance. The generated predictions serve a dual role: they act as historical appearance context for subsequent autoregressive predictions, while simultaneously learning temporal evolution patterns of target appearances, thus capturing the underlying sequential dependencies of object deformation.

3.2. Trajectory Verification

Beyond appearance prediction, which aims to capture appearance variations across continuous video frames, the physical properties of the target's trajectory are overlooked. These properties, however, are critical for robust tracking [59], particularly in challenging scenarios involving occlusions or distractors. Thus, we introduce trajectory verification. Our approach leverages the predicted appearance as templates to track the target back to its initial state in the reverse video sequence. This process ensures both the consistency of the target's trajectory and the preservation of its identity throughout successive tracking.

Our trajectory verification can be regarded as tracking in an rewind autoregressive manner, the trajectory generated preceding, prompts to verification subsequent motion evolution, formulated as:

$$P\left(\boldsymbol{Y}^{t} \middle| \underbrace{\overleftarrow{\boldsymbol{Y}^{t+N:t+1}, \hat{\boldsymbol{Z}}^{t+N:t+1}}}_{\text{reversed trajectory appearance}}, \boldsymbol{X}^{t}\right) \tag{2}$$

where $\hat{Z}^{t+N:t+1}$ is derived from appearance predictions $\hat{X}^{t+N:t+1}$ through spatial transformation to generate centered object templates, $\boldsymbol{Y}^{t}$ denotes the coordinates to be back-tracked at time step t, $\boldsymbol{Y}^{t+N:t+1}$ represents the trajectory in reverse, and $\leftarrow$ represent the sequence order (from $t+N$ to $t+1$), which is reverse to Eq 1.

The autoregressive trajectory verification opts for ViT same as appearance prediction. Initially, both the predicted appearance (used as templates) and search images are divided into patches, flattened, and projected into a sequence of visual embeddings. Similar to ARTrack, we map the object's trajectory across frames to a common coordinate system and tokenize them as trajectory prompts, using a shared vocabulary [11, 12]. Next, we concatenate visual tokens, trajectory tokens, and four command tokens (each corresponding to a bounding box token) to form the input sequence. This sequence is fed into the ViT to generate the current coordinates. Operating autoregressively, the model processes the video sequence backward from the last frame to the first. The generated trajectory in the first frame is trained to match the initialized target state, to verify the motion consistency in continuous video.

3.3. Pretraining Augmentation

In our autoregressive generative pretraining, we initially follow the conventional tracking approach by constraining the target to the center of the search region [2]. However, we

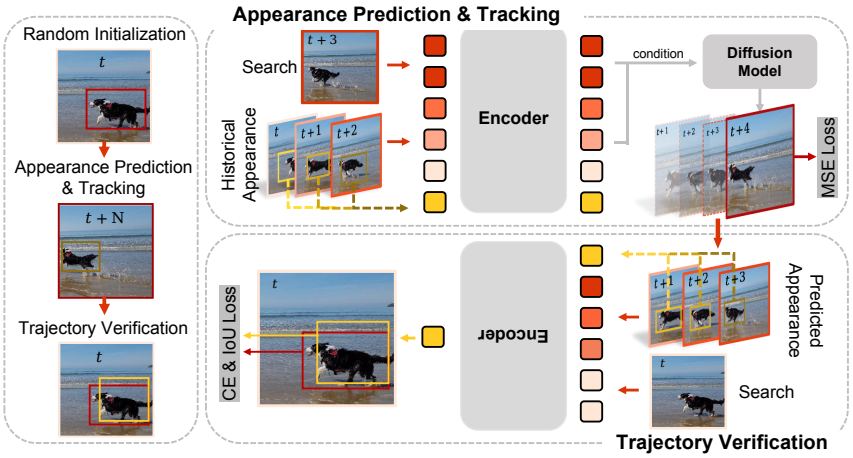


Figure 2. **Pipeline of ARP**. For appearance prediction, the search image and historical appearance are first fed into an encoder for feature extraction, after which a diffusion model predicts the target's appearance in the next-frame. For trajectory verification, the same encoder utilizes the generated appearance as templates to back-track the target's trajectory in reverse order.

observe that this center-biased strategy forces the encoder overly favor to generate bounding boxes near the center ever, which hinders the model's ability to learn motion consistency.

To alleviate the above limitation, we introduce random spatial perturbations during trajectory verification by manipulating the search images. Specifically, we apply two types of transformations: random translations of the crop center and scale variations of the target object. These perturbations create challenging trajectory patterns that boost the model's ability to capture motion dependencies between consecutive frames.

Beyond that, through our analysis, we observed that two key factors can significantly degrade pretraining performance: (1) inappropriate scale variations, particularly extreme aspect ratios, and (2) suboptimal target positions where objects are located outside the field of view. To address these issues, we implement a set of carefully calibrated constraints on random initial bounding boxes. These constraints ensure that the bounding box size remains suitable (neither extremely large nor small), while keeping the initialization box fully visible within the first frame. This calibration strategy effectively preserves the spatial integrity of the target objects during pretraining.

3.4. Optimization Strategy

Our training framework incorporates a hybrid loss function that jointly optimizes appearance prediction and trajectory verification. For appearance evolving, we employ the mean squared error (MSE) to minimize the discrepancy between the generated appearance and the ground truth appearance of the next frame. The trajectory verification is governed by a structured objective that combines two complementary components: a sequence-based log-likelihood term that captures temporal dependencies in coordinate trajectories, and a task-agnostic IoU-based geometric constraint [28] that ensures spatial correlation. The comprehensive optimization objective is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_{\text{iou}}\mathcal{L}_{\text{iou}} + \lambda_{\text{mse}}\mathcal{L}_{\text{mse}} \tag{3}$$

where $\mathcal{L}_{\text{ce}}$ represents the coordinate generative cross-entropy loss, $\mathcal{L}_{\text{iou}}$ is the SIoU loss, and $\mathcal{L}_{\text{mse}}$ is the MSE loss, quantifies appearance prediction quality. The coefficients λ_{iou} and λ_{mse} balance the contributions of different loss terms.

4. Experiments

4.1. Implementation Details

Pretraining Datasets. We train our ARP on multiple video datasets, including VastTrack [49], Kinetics-400 (K400) [37], Kinetics-600 (K600) [7], and Kinetics-700 (K700) [8]. VastTrack is a comprehensive tracking benchmark focusing on vast object categories and complex scenarios, while Kinetics is a series of datasets dedicated to temporal action recognition with rich motion evolution and appearance dynamic, which is coupled with our ARP.

Training Strategy. The training process spans 600 epochs, with frames randomly sampled from each video using a maximum interval of 50 frames. We employ a progressive learning scheme where the clip length incrementally increases from 2 to 7 frames at every 100 epochs, enabling the model to capture increasingly complex temporal dependencies. For model-specific configurations, detailed training strategies are provided in the appendix.

Fine-tuning. We incorporate our pretrained encoder into the ARTrackV2 framework to construct ARPTrack. For a

Methods	GOT-10k*			LaSOT			LaSOText			TrackingNet		
	$AO(\%)$	$SR_{0.5}(\%)$	$SR_{0.75}(\%)$	$AUC(\%)$	$P_{Norm}(\%)$	$P(\%)$	$AUC(\%)$	$P_{Norm}(\%)$	$P(\%)$	$AUC(\%)$	$P_{Norm}(\%)$	$P(\%)$
SiamFC$_{255}$ [2]	34.8	35.3	9.8	33.6	42.0	33.9	23.0	31.1	26.9	57.1	66.3	53.3
ECO$_{224}$ [19]	31.6	30.9	11.1	32.4	33.8	30.1	22.0	25.2	24.0	55.4	61.8	49.2
DiMP$_{288}$ [46]	61.1	71.7	49.2	56.9	65.0	56.7	39.2	47.6	45.1	74.0	80.1	68.7
SiamR-CNN$_{255}$ [56]	64.9	72.8	59.7	64.8	72.2	-	-	-	-	81.2	85.4	80.0
Ocean$_{255}$ [69]	61.1	72.1	47.3	56.0	65.1	56.6	-	-	-	-	-	-
TrDiMP$_{352}$ [58]	67.1	77.7	58.3	63.9	-	61.4	-	-	-	78.4	83.3	73.1
SLT-TrDiMP$_{352}$ [39]	67.5	78.8	58.7	64.4	73.5	-	-	-	-	78.1	83.1	73.1
TransT$_{256}$ [13]	67.1	76.8	60.9	64.9	73.8	69.0	-	-	-	81.4	86.7	80.3
STARK$_{320}$ [63]	68.8	78.1	64.1	67.1	77.0	-	-	-	-	82.0	86.9	-
SwinTrack-B$_{384}$ [44]	72.4	80.5	67.8	71.3	-	76.5	49.1	-	55.6	84.0	-	82.8
MixFormer-L$_{320}$ [15]	-	-	-	70.1	79.9	76.3	-	-	-	83.9	88.9	83.1
OSTrack$_{384}$ [66]	73.7	83.2	70.8	71.1	81.1	77.6	50.5	61.3	57.6	83.9	88.5	83.2
CTTrack-L$_{320}$ [54]	72.8	81.3	71.5	69.8	79.7	76.2	-	-	-	84.9	89.1	83.5
GRM-B$_{256}$ [27]	73.4	82.9	70.4	69.9	79.3	75.8	-	-	-	84.0	88.7	83.3
GRM-L$_{320}$ [27]	-	-	-	71.4	81.2	77.9	-	-	-	84.4	88.9	84.0
SeqTrack-B$_{256}$ [14]	74.7	84.7	71.8	69.9	79.7	76.3	49.5	60.8	56.3	83.3	88.3	82.2
SeqTrack-L$_{384}$ [14]	74.8	81.9	72.2	72.5	81.5	79.3	50.7	61.6	57.5	85.5	89.8	85.8
ARTrack$_{256}$ [59]	73.5	82.2	70.9	70.4	79.5	76.6	46.4	56.5	52.3	84.2	88.7	83.5
ARTrack-L$_{384}$ [59]	78.5	87.4	77.8	73.1	82.2	80.3	52.8	62.9	59.7	85.6	89.6	86.0
HIPTrack$_{384}$ [6]	77.4	88.0	74.5	72.7	82.9	79.5	-	-	-	84.5	89.1	83.8
DiffusionTrack-B$_{256}$ [61]	74.8	85.4	72.0	70.8	79.8	76.7	-	-	-	83.8	88.2	82.1
DiffusionTrack-L$_{256}$ [61]	74.7	85.6	71.8	72.3	81.8	79.1	-	-	-	85.2	89.6	84.8
AQATrack$_{256}$ [62]	73.8	83.2	72.1	71.4	81.9	78.6	51.2	62.2	58.9	83.8	88.6	83.1
AQATrack$_{384}$ [62]	76.0	85.2	74.9	72.7	82.9	80.2	52.7	64.2	60.8	84.8	89.3	84.3
ARTrackV2$_{256}$ [1]	75.9	85.4	72.7	71.6	80.2	77.2	50.8	61.9	57.7	84.9	89.3	84.5
ARTrackV2-L$_{384}$ [1]	79.5	87.8	79.6	73.6	82.8	81.1	53.4	63.7	60.2	86.1	90.4	86.2
ARPTrack$_{256}$	77.7	87.3	74.3	72.6	81.4	78.5	52.0	62.9	58.7	85.5	90.0	85.3
ARPTrack-L$_{384}$	**81.5**	**90.6**	**80.5**	**74.2**	**83.4**	**81.7**	**54.2**	**64.4**	**61.2**	**86.6**	**91.1**	**87.4**

Table 1. State-of-the-art comparison on GOT-10k, TrackingNet, LaSOT and LaSOText. Where * denotes only trained on GOT-10k. The number in the subscript denotes the search region resolution. Best in **bold**, second best underlined, and third best underwave.

fair comparison, we follow the same fine-tuning protocols as ARTrackV2. The training data consists of three benchmark datasets: GOT-10k [35] (excluding 1k sequences from its training split as per STARK [63]), TrackingNet [47], and LaSOT [24]. For the evaluation of the GOT-10k test set, we strictly follow its one-shot protocol by training our models on the complete GOT-10k training split. All hyperparameters, including learning rate, number of epochs, and weight decay, remain identical to ARTrackV2 settings to ensure a fair comparison, for two-stage training pipeline.

Model Variants and Architecture. We implement two variants of our ARP tracker:

- **ARPTrack$_{256}$:** Built on ViT-Base backbone with template and search size of [256×256]
- **ARPTrack-L$_{384}$:** Utilizing ViT-Large backbone with template and search size of [384×384]

4.2. State-of-the-art Comparison

We evaluate our ARPTrack$_{256}$ and ARPTrack-L$_{384}$ (both are pretrained on K700) against state-of-the-art approaches on multiple benchmarks, the results are in Table 1:

GOT-10k. GOT-10k [35] presents a challenging one-shot learning scenario where training and testing object categories are deliberately separated. Our ARPTrack$_{256}$ significantly outperforms other trackers with the same resolution, while our ARPTrack-L$_{384}$ achieves the state-of-the-art performance, surpassing the previous state-of-the-art ARTrackV2-L$_{384}$ by 2.0% in AO and 2.8% in $SR_{0.5}$.

LaSOT. LaSOT [24] serves as a crucial benchmark for evaluating long-term tracking capabilities, comprising 280 diverse test videos with extended sequences. On this challenging dataset, ARPTrack-L$_{384}$ achieves 74.2% AUC. The model also demonstrates superior precision and normalized precision, establishing new state-of-the-art performance across all metrics.

LaSOText. LaSOText [25] serves as an extension of LaSOT, including an additional 150 videos. These new sequences introduce challenging tracking scenarios, such as occlusions and variations in small objects. On this dataset, ARPTrack-L$_{384}$ achieves 54.2% AUC, 64.4% precision, and 61.2% normalized precision, validating our model's robustness in handling complex scenarios.

TrackingNet. TrackingNet [47] represents a large-scale tracking benchmark comprising over 30,000 sequences that encompass diverse real-world scenarios. On it's diverse test

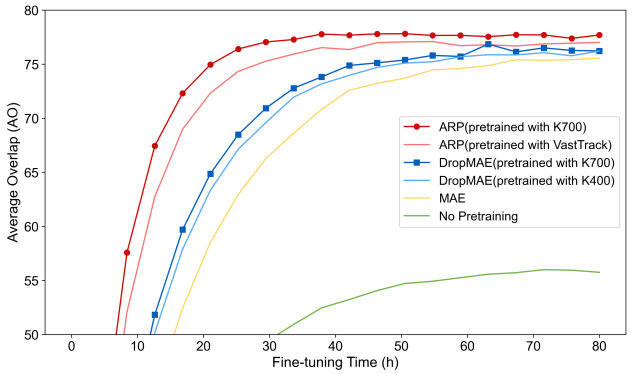


Figure 3. Comparison with **different pretraining approaches**.

set of 511 sequences, ARPTrack-L_{384} achieves 86.6% AUC, 91.1% precision, and 87.4% normalized precision, confirming its superior performance across diverse tracking conditions.

4.3. Experimental Analyses

We conduct comprehensive experiments to examine the key characteristics of ARP. Unless otherwise specified, all analyses are performed using our ARP model trained on VastTrack, with evaluations following the GOT-10k [35] test protocol for fine-tuning. Default configurations are highlighted in gray.

Different Pretraining Approaches. To compare our approach with existing methods, we conduct experiments on both image-level (MAE) and pairs-level (DropMAE) pretraining models. To evaluate the performance change over the fine-tuning time, we integrate all approaches only on ARTrackV2 first stage, with results presented in Figure 3. ARP and DropMAE are pretrained on videos, while MAE uses ImageNet-1K. All models employ the same fine-tuning strategy for fair comparison. Results show that both video-specific approaches (ARP and DropMAE) significantly outperform MAE and the no-pretraining baseline. While DropMAE enhances performance through spatial-attention dropout between paired frames, it remains limited in modeling long-rang successive temporal dependencies. Our approach outperforms DropMAE even when trained on a smaller dataset (VastTrack) compared to DropMAE's training on the larger K700 dataset. This performance advantage demonstrates our model's superior capability in capturing both object deformation and motion essential for tracking tasks. Notably, ARP converges in about 20 hours during fine-tuning, compared to more than 30 hours for other methods, indicating that ARP enables both better representations and more efficient transfer learning ability.

Scaling-up Ability. The success of large language models is inseparable from their ability to learn from large-scale data. Inspired by this, we investigate our model's performance across datasets of varying scales. We conduct experiments on four datasets, as shown in Table 2. While our ARP achieves competitive results on VastTrack, it demonstrates superior performance on Kinetics series, which features rich deformable and dynamic patterns.

Furthermore, when scaling up to larger datasets (from K400 to K700), we observe consistent performance improvements, ultimately achieving 77.7% AO when trained on K700. These results validate our model's strong scaling capability, indicating that ARP can effectively learn more robust patterns of object appearance evolution and motion dynamics from larger datasets. This scalability demonstrates the potential of our pretrained model to leverage increasingly extensive datasets in the future.

Datasets	Videos	Categories	Pretrain. Time (d)	AO	$SR_{0.5}$	$SR_{0.75}$
VastTrack	50,610	2,115	3.9	76.8	86.2	73.5
K400	246,535	400	14.7	77.2	86.9	73.7
K600	392,622	600	23.4	77.4	87.1	73.9
K700	545,317	700	29.3	**77.7**	**87.3**	**74.3**

Table 2. Comparison on **different-scale datasets**. The pretraining time is measured on 8 NVIDIA A6000 GPUs

Tracker	Pretraining	AO	$SR_{0.5}$	$SR_{0.75}$
OSTrack_{256}	MAE	71.0	80.4	68.2
	ARP	72.5	82.5	70.2
ARTrack_{256}	MAE	73.5	82.2	70.9
	ARP	75.2	84.3	72.5
ARTrackV2_{256}	MAE	75.9	85.4	72.7
	ARP	**76.8**	**86.2**	**73.5**

Table 3. **Generalization capability** of ARP on different trackers.

Generalization Capability. To demonstrate ARP's ability to generalize across different trackers, we integrate it into various tracking frameworks and compare their performance against MAE pretraining, as shown in Table 3. All trackers equipped with ARP exhibit significant performance improvements compared to their MAE-pretrained counterparts. These results validate that ARP's effectiveness is not only limited to sequential generation trackers (ARTrack and ARTrackV2) but also extends to per-frame template matching trackers (OSTrack), which emphasize the generalization capability of ARP.

Appearance	Trajectory	AO	$SR_{0.5}$	$SR_{0.75}$
	✓	76.1	85.3	72.4
✓		76.3	85.7	72.9
✓	✓	**76.8**	**86.2**	**73.5**

Table 4. Comparative analysis of **key components**.

Components Analysis. ARP models both object deformation and motion dynamics in videos through two key components: appearance prediction and trajectory verification. We evaluate the synergistic effect of appearance prediction and trajectory verification by comparing the performance between appearance prediction alone and our complete model. As shown in Table 4, while appearance prediction alone achieves 76.3% AO, incorporating trajectory verification further improves the performance by 0.5%. These results highlight the complementary roles of both components: appearance prediction generates robust visual features and templates for tracking guidance, while trajectory verification ensures temporal consistency in motion patterns.

Length	Fixed-length			Progressive-length		
	AO	$SR_{0.5}$	$SR_{0.75}$	AO	$SR_{0.5}$	$SR_{0.75}$
3	74.6	84.3	72.3	73.7	83.3	71.6
5	76.1	85.4	72.7	76.3	85.7	73.1
7	76.5	85.8	73.2	**76.8**	**86.2**	**73.5**
9	76.4	85.4	73.2	76.7	85.9	73.3

Table 5. Impact of **sequence length strategies** on tracking performance. Every setting trains for the same epochs.

Sequence Length. We sample video clip sequences for training to maintain temporal consistency, where the sequence length directly influences the model's ability to learn temporal dependencies. To explore the sequence length on tracking performance, we conduct experiments with varying lengths of 3, 5, 7, and 9. Additionally, we investigate two training strategies: fixed-length and progressive-length. In the fixed-length approach, we maintain a consistent sequence length throughout training, while the progressive-length strategy incrementally increases the sequence length from 2 to each target maximum (3, 5, 7, or 9), with training epochs equally allocated across all intermediate lengths.

As shown in Table 5, both strategies yield consistent performance improvements up to sequence length 7, but exhibit a slight decrease at length 9. This suggests that while increasing the sequence length helps obtain more temporal information, excessive length introduces unnecessary interference, such as irrelevant motion dynamics or appearance variations for current frame, thus degrading performance. Regarding training strategies, the fixed-length approach performs better when the sequence length is short (length=3), but the progressive strategy shows superior performance starting from sequence length 5. This is because short sequences in progressive training cannot adequately capture temporal dependencies, but with longer sequences, progressive training effectively captures both short-range and long-range temporal dependencies.

Appearance Generation Method. In ARP, we employ a conditional diffusion model to predict the next-frame appearance. This approach differs from MAE, a successful pretraining method that also relies on frame generation. MAE employs a mask strategy to learn representations by reconstructing masked regions of the current frame. In this section, we compare these two generation methods and evaluate different generation targets (current versus future frame appearances).

Module	Frame	AO	$SR_{0.5}$	$SR_{0.75}$
mask	current	76.0	84.9	72.1
diffusion		76.1	85.5	72.8
mask	future	76.4	85.9	73.4
diffusion		**76.8**	**86.2**	**73.5**

Table 6. Comparisons of **appearance generation model**.

As shown in Table 6, future frame generation consistently outperforms current frame generation, indicating that learning temporal-aware representations better serves video tracking tasks. Moreover, in future frame generation, the conditional diffusion model achieves superior performance compared to mask reconstruction (MAE), demonstrating the diffusion model's effectiveness in modeling progressive appearance deformations.

Pretraining Augmentation	AO	$SR_{0.5}$	$SR_{0.75}$
skratch	-	-	-
+ size constraints	60.4	65.3	44.7
+ position constraints	73.9	82.7	70.9
+ center jitter	76.2	85.6	72.9
+ scale jitter	**76.8**	**86.2**	**73.5**

Table 7. Effect of **pretraining augmentation**.

Pretraining Augmentation. During training, we employ spatial perturbations and bounding box constraints to enhance model robustness. Specifically, we introduce scale jitter to control random variations in image size, and center jitter to regulate random offsets from the original center position in search regions. To ensure training feasibility, we implement constraints on the initial bounding box to maintain a suitable size and full visibility in the first frame. As shown in Table 7, both center and scale jitter improve the model's stability. Without translations in the search region, the model may lead to ineffective training by forcing the encoder to overly favor generating bounding boxes near the center, thus demonstrating the importance of these augmentation strategies.

Computational Cost Metrics. As shown in Table 8, we conduct a comprehensive evaluation of computational efficiency by analyzing memory usage, FLOPs, and inference speed. ARPTrack shares the same architecture and FLOPs

Tracker	AO	Memory Usage	FLOPs	FPS
ARPTrack_{256}	77.7	1536MiB	51G	91
ARTrackV2_{256}	75.9	1536MiB	51G	91
SeqTrack_{256}	74.7	1332MiB	66G	42
MixViT_{288}	72.7	2180MiB	43G	83

Table 8. **Computational cost metrics** of ARPTrack.

as ARTrackV2 but achieves competitive performance. Notably, our method demonstrates superior efficiency compared to SeqTrack with reduced FLOPs and higher inference speed. Although MixViT consumes fewer FLOPs, it requires substantially more memory and runs at a lower frame rate. These results validate that ARPTrack strikes an optimal balance between tracking accuracy and computational efficiency, making it particularly suitable for practical applications.

Visualization. To comprehensively evaluate our method's effectiveness in appearance and trajectory prediction, we first compare our predicted appearance with ground truth appearances across various challenging scenarios, as illustrated in Figure 4. These scenarios include partial occlusions, appearance variations, interactive distractors, and out-of-view targets. The experimental results demonstrate that our method successfully captures object deformation patterns based on historical information.

Additionally, we visualize the cross-attention maps between appearance tokens and search regions, focusing particularly on challenging cases such as motion blur and other complex scenarios, as illustrated in Figure 5. Notably, even when the initial bounding box only partially contains the target object, our model maintains attention on the coherent target throughout subsequent frames. The visualization results demonstrate that our model effectively captures the evolution of trajectory features across consecutive frames.

Conventional trackers struggle with occluded or out-of-view targets due to lost contextual information. In contrast, our model maintains robust tracking by leveraging temporal dependencies to model appearance deformation and motion patterns.

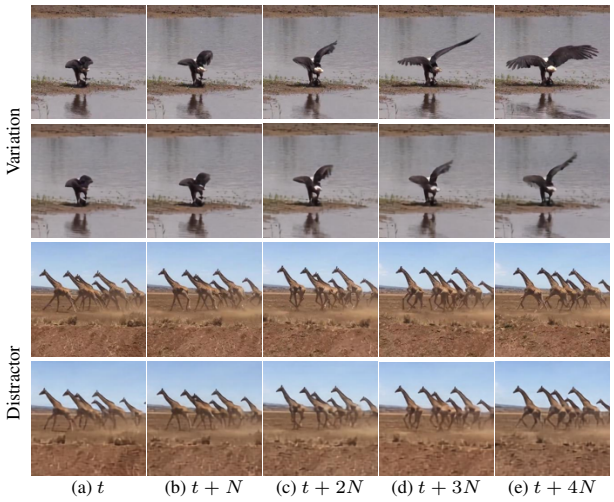


Figure 4. **Appearance prediction visualization**. For each video sequence, we show the ground truth appearance (top) and predicted appearance by ARP. (a)-(e) represented sampling video frames at a fixed interval of N.

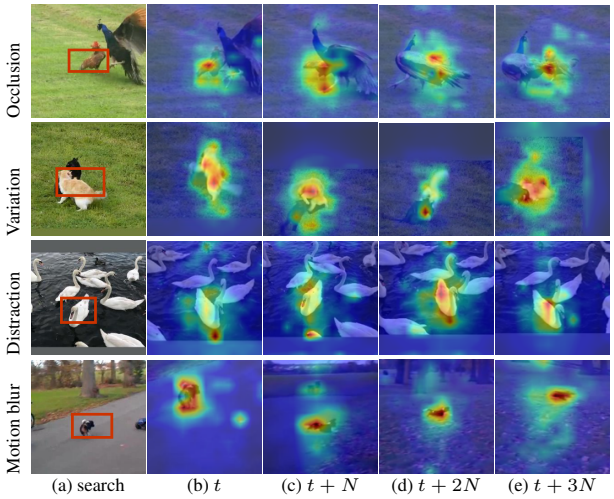


Figure 5. **Attention visualization**. (a): Search region and template. The red boxes denote the randomly initialized bounding box. (b)-(e): Cross-attention map during trajectory vitrifaction of ARP.

5. Conclusion

We propose ARP, an unsupervised autoregressive pretraining that challenges the widely adopted image-level pretrained backbones in prevailing trackers, which fail to capture dynamics in successive videos critical for tracking. ARP incorporates appearance prediction and trajectory verification, encouraging the encoder to learn object deformation and motion pattern, aka temporal dependencies, for consistency tracking. We further integrate ARP into ARTrackV2, which achieves SoTA performance across multiple benchmarks, demonstrating that video-level pretraining better aligns with the nature of tracking. Beyond that, we envision ARP to provide new insights for a broader range of video-level tasks, such as video segmentation.

Future Work. Our ARP can be further enhanced by incorporating larger web datasets, more powerful models (e.g., ViT-G), and higher spatial resolution input videos. Beyond that, ARP has so far only been evaluated on tracking tasks, which rely on the coordinate-based representation of bounding boxes. Its effectiveness in other tasks that involve temporal dependencies, such as video segmentation, remains to be explored.

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References

[1] Yifan Bai, Zeyang Zhao, Yihong Gong, and Xing Wei. Artrackv2: Prompting autoregressive tracker where to look and how to describe. In CVPR, 2024. 1, 2, 5

[2] Luca Bertinetto, Jack Valmadre, Joao F Henriques, Andrea Vedaldi, and Philip HS Torr. Fully-convolutional siamese networks for object tracking. In ECCV, 2016. 1, 2, 3, 5

[3] Goutam Bhat, Joakim Johnander, Martin Danelljan, Fahad Shahbaz Khan, and Michael Felsberg. Unveiling the power of deep tracking. In ECCV, 2018. 1

[4] Goutam Bhat, Martin Danelljan, Luc Van Gool, and Radu Timofte. Learning discriminative model prediction for tracking. In ICCV, 2019. 2

[5] Tom B Brown. Language models are few-shot learners. arXiv, 2020. 2, 3

[6] Wenrui Cai, Qingjie Liu, and Yunhong Wang. Hiptrack: Visual tracking with historical prompts. In CVPR, 2024. 1, 5

[7] Joao Carreira, Eric Noland, Andras Banki-Horvath, Chloe Hillier, and Andrew Zisserman. A short note about kinetics-600. arXiv, 2018. 4

[8] Joao Carreira, Eric Noland, Chloe Hillier, and Andrew Zisserman. A short note on the kinetics-700 human action dataset. arXiv, 2019. 4

[9] Boyu Chen, Peixia Li, Lei Bai, Lei Qiao, Qiuhong Shen, Bo Li, Weihao Gan, Wei Wu, and Wanli Ouyang. Backbone is all your need: A simplified architecture for visual object tracking. In ECCV. Springer, 2022. 2

[10] Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Heewoo Jun, David Luan, and Ilya Sutskever. Generative pretraining from pixels. In ICML, 2020. 3

[11] Ting Chen, Saurabh Saxena, Lala Li, David J Fleet, and Geoffrey Hinton. Pix2seq: A language modeling framework for object detection. ICLR, 2021. 2, 3

[12] Ting Chen, Saurabh Saxena, Lala Li, Tsung-Yi Lin, David J Fleet, and Geoffrey Hinton. A unified sequence interface for vision tasks. In NeurIPS, 2022. 3

[13] Xin Chen, Bin Yan, Jiawen Zhu, Dong Wang, Xiaoyun Yang, and Huchuan Lu. Transformer tracking. In CVPR, 2021. 2, 5

[14] Xin Chen, Houwen Peng, Dong Wang, Huchuan Lu, and Han Hu. Seqtrack: Sequence to sequence learning for visual object tracking. In CVPR, 2023. 1, 2, 5

[15] Yutao Cui, Cheng Jiang, Limin Wang, and Gangshan Wu. Mixformer: End-to-end tracking with iterative mixed attention. In CVPR, 2022. 2, 5

[16] Yutao Cui, Tianhui Song, Gangshan Wu, and Limin Wang. Mixformerv2: Efficient fully transformer tracking. NeurIPS, 2023. 2

[17] Yutao Cui, Cheng Jiang, Gangshan Wu, and Limin Wang. Mixformer: End-to-end tracking with iterative mixed attention. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2024. 2

[18] Kenan Dai, Yunhua Zhang, Dong Wang, Jianhua Li, Huchuan Lu, and Xiaoyun Yang. High-performance long-term tracking with meta-updater. In CVPR, 2020. 2

[19] Martin Danelljan, Goutam Bhat, Fahad Shahbaz Khan, and Michael Felsberg. Eco: Efficient convolution operators for tracking. In CVPR, 2017. 2, 5

[20] Martin Danelljan, Goutam Bhat, Fahad Shahbaz Khan, and Michael Felsberg. Atom: Accurate tracking by overlap maximization. In CVPR, 2019. 2

[21] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In ICLR, 2021. 3

[22] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. arXiv, 2024. 2, 3

[23] Heng Fan and Haibin Ling. Siamese cascaded region proposal networks for real-time visual tracking. In CVPR, 2019. 1, 2

[24] Heng Fan, Liting Lin, Fan Yang, Peng Chu, Ge Deng, Sijia Yu, Hexin Bai, Yong Xu, Chunyuan Liao, and Haibin Ling. Lasot: A high-quality benchmark for large-scale single object tracking. In CVPR, 2019. 5

[25] Heng Fan, Hexin Bai, Liting Lin, Fan Yang, Peng Chu, Ge Deng, Sijia Yu, Mingzhen Huang, Juehuan Liu, Yong Xu, Chunyuan Liao, Lin Yuan, and Haibin Ling. Lasot: A high-quality large-scale single object tracking benchmark. International Journal of Computer Vision, 2021. 5

[26] Shenyuan Gao, Chunluan Zhou, Chao Ma, Xinggang Wang, and Junsong Yuan. Aiatrack: Attention in attention for transformer visual tracking. In European Conference on Computer Vision, 2022. 2

[27] Shenyuan Gao, Chunluan Zhou, and Jun Zhang. Generalized relation modeling for transformer tracking. In CVPR, 2023. 2, 5

[28] Zhora Gevorgyan. Siou loss: More powerful learning for bounding box regression. arXiv, 2022. 4

[29] Dongyan Guo, Jun Wang, Ying Cui, Zhenhua Wang, and Shengyong Chen. Siamcar: Siamese fully convolutional classification and regression for visual tracking. In CVPR, 2020. 1, 2

[30] Agrim Gupta, Jiajun Wu, Jia Deng, and Li Fei-Fei. Siamese masked autoencoders. arXiv, 2023. 2

[31] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In CVPR, 2020. 2

[32] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In CVPR, 2022. 2

[33] Kaijie He, Canlong Zhang, Sheng Xie, Zhixin Li, and Zhiwen Wang. Target-aware tracking with long-term context attention. AAAI, 2023. 2

[34] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. NIPS, 2020. 3

[35] Lianghua Huang, Xin Zhao, and Kaiqi Huang. Got-10k: A large high-diversity benchmark for generic object tracking in the wild. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2019. 5, 6

[36] Sajid Javed, Martin Danelljan, Fahad Shahbaz Khan, Muhammad Haris Khan, Michael Felsberg, and Jiri Matas. Visual object tracking with discriminative filters and siamese networks: a survey and outlook. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022. 1

[37] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. The kinetics human action video dataset. arXiv, 2017. 4

[38] Hamed Kiani Galoogahi, Ashton Fagg, and Simon Lucey. Learning background-aware correlation filters for visual tracking. In ICCV, 2017. 1

[39] Minji Kim, Seungkwan Lee, Jungseul Ok, Bohyung Han, and Minsu Cho. Towards sequence-level training for visual tracking. In ECCV, 2022. 1, 5

[40] Bo Li, Junjie Yan, Wei Wu, Zheng Zhu, and Xiaolin Hu. High performance visual tracking with siamese region proposal network. 2018. 2

[41] Bo Li, Wei Wu, Qiang Wang, Fangyi Zhang, Junliang Xing, and Junjie Yan. Siamrpn++: Evolution of siamese visual tracking with very deep networks. In CVPR, 2019. 1, 2

[42] Feng Li, Cheng Tian, Wangmeng Zuo, Lei Zhang, and Ming-Hsuan Yang. Learning spatial-temporal regularized correlation filters for visual tracking. In CVPR, 2018. 1

[43] Peixia Li, Boyu Chen, Wanli Ouyang, Dong Wang, Xiaoyun Yang, and Huchuan Lu. Gradnet: Gradient-guided network for visual object tracking. In ICCV, 2019. 2

[44] Liting Lin, Heng Fan, Yong Xu, and Haibin Ling. Swintrack: A simple and strong baseline for transformer tracking. In NeurIPS, 2022. 1, 2, 5

[45] R.D. Lin, Pengcheng Weng, Yinqiao Wang, Han Ding, Jinsong Han, and Fei Wang. Hilots: High-low temporal sensitive representation learning for semi-supervised lidar segmentation in autonomous driving. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025. 2

[46] Alan Lukežic, Tomáš Vojír, Luka Cehovin Zajc, Jirí Matas, and Matej Kristan. Discriminative correlation filter with channel and spatial reliability. In CVPR, 2017. 2, 5

[47] Matthias Muller, Adel Bibi, Silvio Giancola, Salman Alsubaihi, and Bernard Ghanem. Trackingnet: A large-scale dataset and benchmark for object tracking in the wild. In ECCV, 2018. 5

[48] Hyeonseob Nam and Bohyung Han. Learning multi-domain convolutional neural networks for visual tracking. In CVPR, 2016. 1

[49] Liang Peng, Junyuan Gao, Xinran Liu, Weihong Li, Shaohua Dong, Zhipeng Zhang, Heng Fan, and Libo Zhang. Vastrack: Vast category visual object tracking. arXiv, 2024. 4

[50] Alec Radford. Improving language understanding by generative pre-training. 2018. 2, 3

[51] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. OpenAI blog, 2019. 2, 3

[52] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. JMLR, 2020. 2

[53] Zikai Song, Junqing Yu, Yi-Ping Phoebe Chen, and Wei Yang. Transformer tracking with cyclic shifting window attention. In CVPR, 2022. 2

[54] Zikai Song, Run Luo, Junqing Yu, Yi-Ping Phoebe Chen, and Wei Yang. Compact transformer tracker with correlative masked modeling. AAAI, 2023. 2, 5

[55] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. arXiv, 2023. 2, 3

[56] Paul Voigtlaender, Jonathon Luiten, Philip HS Torr, and Bastian Leibe. Siam r-cnn: Visual tracking by re-detection. In CVPR, 2020. 1, 2, 5

[57] Guangting Wang, Chong Luo, Xiaoyan Sun, Zhiwei Xiong, and Wenjun Zeng. Tracking by instance detection: A meta-learning approach. In CVPR, 2020. 2

[58] Ning Wang, Wengang Zhou, Jie Wang, and Houqiang Li. Transformer meets tracker: Exploiting temporal context for robust visual tracking. In CVPR, 2021. 2, 5

[59] Xing Wei, Yifan Bai, Yongchao Zheng, Dahu Shi, and Yihong Gong. Autoregressive visual tracking. In CVPR, 2023. 1, 2, 3, 5

[60] Qiangqiang Wu, Tianyu Yang, Ziquan Liu, Baoyuan Wu, Ying Shan, and Antoni B Chan. Dropmae: Masked autoencoders with spatial-attention dropout for tracking tasks. In CVPR, 2023. 2

[61] Fei Xie, Zhongdao Wang, and Chao Ma. Diffusiontrack: Point set diffusion model for visual object tracking. In CVPR, 2024. 5

[62] Jinxia Xie, Bineng Zhong, Zhiyi Mo, Shengping Zhang, Liangtao Shi, Shuxiang Song, and Rongrong Ji. Autoregressive queries for adaptive tracking with spatio-temporal transformers. In CVPR, 2024. 1, 5

[63] Bin Yan, Houwen Peng, Jianlong Fu, Dong Wang, and Huchuan Lu. Learning spatio-temporal transformer for visual tracking. In ICCV, 2021. 2, 5

[64] Tianyu Yang and Antoni B Chan. Learning dynamic memory networks for object tracking. In ECCV, 2018. 2

[65] Tianyu Yang, Pengfei Xu, Runbo Hu, Hua Chai, and Antoni B Chan. Roam: Recurrently optimizing tracking model. In CVPR, 2020. 2

[66] Botao Ye, Hong Chang, Bingpeng Ma, and Shiguang Shan. Joint feature learning and relation modeling for tracking: A one-stream framework. In ECCV, 2022. 1, 2, 5

[67] Lichao Zhang, Abel Gonzalez-Garcia, Joost Van De Weijer, Martin Danelljan, and Fahad Shahbaz Khan. Learning the model update for siamese trackers. In ICCV, 2019. 2

[68] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In ICCV, 2023. 3

[69] Zhipeng Zhang, Houwen Peng, Jianlong Fu, Bing Li, and Weiming Hu. Ocean: Object-aware anchor-free tracking. In ECCV, 2020. 2, 5

[70] Yaozong Zheng, Bineng Zhong, Qihua Liang, Zhiyi Mo, Shengping Zhang, and Xianxian Li. Odtrack: Online dense temporal token learning for visual tracking. In AAAI, 2024. 1

[71] Zheng Zhu, Qiang Wang, Bo Li, Wei Wu, Junjie Yan, and Weiming Hu. Distractor-aware siamese networks for visual object tracking. In ECCV, 2018. 2